

Enrolment No: _____ Name of Student: _____

Department/ School: _____

END-TERM MAKE-UP EXAMINATION, EVEN SEMESTER JULY 2024
COURSE CODE - CSET207
MAX. DURATION 2 HRS
COURSE NAME – Computer Networks
PROGRAM – B.Tech
TOTAL MARKS 40

Mapping of Questions to Course and Program Outcomes								
Q.No.	A1	A2	A3	A4	B1	B2	B3	B4
CO	3	1,2	1, 2,3	2, 3	2, 3	2	1, 3	1, 3
PO	3,5,6,7	1,2	1,2,4,5	3, 5, 7, 10, 11	3, 5, 7, 10, 11	1, 3	1, 2, 4, 5	1, 2, 4, 5
BTL*1	2, 3	1, 2, 4	1, 2	3, 4	3, 4	2, 3	2, 3, 4	3, 4

GENERAL INSTRUCTIONS: -

1. Do not write anything on the question paper except **name, enrolment number** and **department/school**.
2. Carrying mobile phones, smartwatches and any other non-permissible materials in the examination hall is an act of **UFM**.

COURSE INSTRUCTIONS:

- a) All the questions are compulsory.
- b) Please write precisely and neatly.
- c) Please make clear diagram wherever required.

SECTION A
Max Marks: 22

A1) Suppose you configure a token bucket with a capacity of 1000 tokens and a token arrival rate of 10 tokens per second. Calculate the time it would take for the token bucket to fill up initially, assuming it starts empty. Additionally, if a packet with a size of 1500 bytes arrives at the token bucket, determine whether it can be transmitted immediately or if it needs to wait for tokens to accumulate. Assume that each token corresponds to one byte of data.

[4 Marks]

A2) Explain the concept of Network Address Translation (NAT) and its significance in IPv4 networking. Additionally, consider the following scenario: A network administrator is tasked with expanding the compressed IPv6 address "**2001:0db8::ff00:42:abcd**" to its full representation. Demonstrate step-by-step how the address is expanded and provide the complete IPv6 address.

[2+3=5 Marks]

A3) Write short notes. Attempt **any four**.

[8 Marks]

1) CSMA/CA

2) UDP vs SCTP

3) Selective Repeat ARQ

4) Active vs Passive FTP Modes

5) PQDN Vs FQDN

6) Socket addressing

A4) As a network administrator in charge of overseeing a corporate network, you frequently manage online training sessions within one of the company's departments. These sessions involve a single presenter communicating with several participants simultaneously. In order to make the best use of network resources, your decision is to introduce multicasting. To ensure proper preparation, it is essential to confirm compatibility with the necessary protocols. When it comes to enabling multicasting for online training sessions, the key protocol you would depend on primarily is, and how does it support efficient communication between a single sender and multiple receivers across the network? **[5 Marks]**

SECTION B

Max Marks: 18

B1) Elaborate upon the requirements of Classless Addressing and Subnetting, also discuss why Classful Addressing failed. Consider a Network Designer with a network block 150.50.0.0/16, is approached by 4 clients. Client A requests 500 Host Addresses, Client B requires 1000 hosts, another 1024 and 512 Hosts are requested by Client C and D respectively. Please pen down the address ranges allotted to the four clients. **[5 Marks]**

B2) You are a network administrator, let us assume you are tasked with analysing the TCP header of a packet captured during a network troubleshooting session. You have been provided with the hexadecimal representation of the TCP header. Your task is to decode the header and compute the values based on its contents. Hexadecimal representation of the TCP header:

$(4500\ 0028\ 0001\ 0000\ 0102\ 0000\ 5006\ 0000)_{16}$

Decode the TCP header and identify the values of the following fields:

a) Source Port

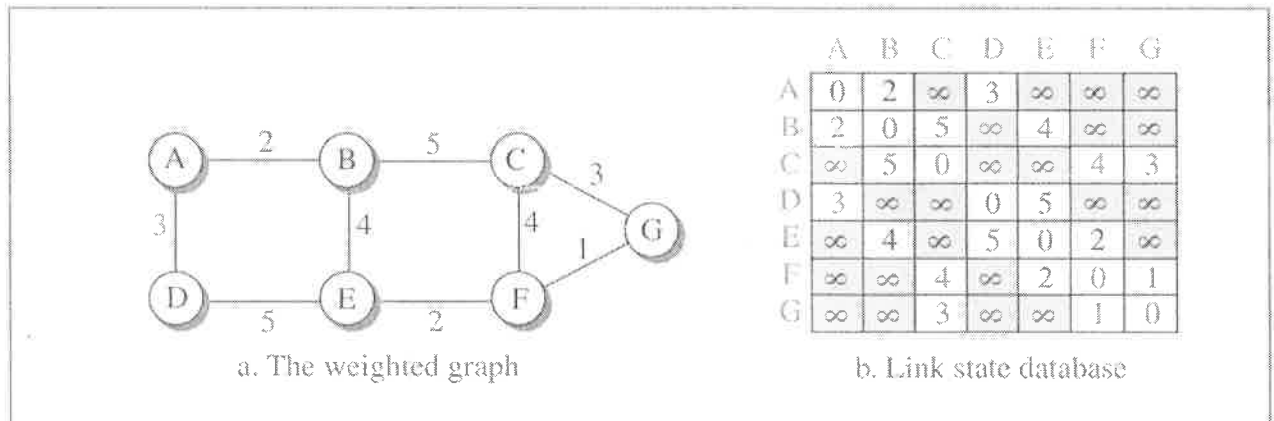
b) Destination Port

c) Sequence Number

d) Acknowledgment Number

[4 Marks]

B3) Consider Bennett University to be an autonomous system with 7 blocks that are positioned at specific distances and may be visited via numerous paths. Furthermore, each block has a network of 100 systems that are connected to a gateway to communicate with other blocks' systems. In the network topology shown below, the gateway of each block is represented as a node, while communication linkages between two blocks are represented as edges. Furthermore, edge metrics and weights are determined by the link's state (throughput or delay). Now you must discover the shortest path tree and routing table while keeping node A as the source node. **[5 Marks]**



B4) Describe fragmentation and its essential role in data communication. Given a scenario where a datagram is divided into 10 fragments and the identification number of the first fragment is 2091, determine the identification number of the last fragment. **[2+2=4 Marks]**

-ALL THE BEST-